Research Readiness Statement

FINESST companion document (1-page limit). NOT anonymized. Red [brackets] are the few items still to confirm.

Degree program. Colton Goodrich is a PhD student in Geosensing Systems Engineering & Sciences at the University of Houston (enrolled August 2025; expected graduation August 2029). Craig L. Glennie (Professor) is the proposal PI.

Research experience. The FI independently designed and built an end-to-end airborne-lidar terrain-analysis pipeline on the Appalachian Plateau (detecting historical disturbance features — excavated pits, access roads, and cleared pads): PDAL point-cloud processing, GDAL/WhiteboxTools bare-earth derivatives, hand-labeled training sets in QGIS, U-Net and instance-segmentation training with honest per-instance evaluation (PyTorch), and DEM-of-Difference change analysis with ICP co-registration and robust (NMAD-based) uncertainty. This is the exact method stack the proposed project uses — every tool named in the S/T/M methodology is one the FI already operates end to end. The FI has additionally applied InSAR-based DEM generation and accuracy assessment against lidar references in support of NASA Surface Topography and Vegetation (STV) objectives — further reinforcing the differencing and uncertainty methods central to this proposal.

Coursework. *Completed:* graduate remote-sensing/geosensing coursework at UH (2025–present) [add course numbers/grades from transcript]; Geospatial Analysis certificate (Weber State, 2019); M.S./B.S. Geology (Brigham Young University). *Planned:* [Bayesian/hierarchical statistical modeling course, term] and [cold-regions geomorphology or hydrology course, or directed reading with the PI, term] — closing the two skill gaps: hierarchical driver modeling (O1) and Antarctic process knowledge.

Skills. Python (NumPy, Pandas, GDAL, Rasterio, geopandas, PyTorch, ArcPy), SQL, MATLAB; PDAL, WhiteboxTools, ArcGIS Pro, QGIS, ENVI; SAR/InSAR, DEM differencing, NMAD/RMSE accuracy assessment; version-controlled, documentation-first practice (public repositories, per-experiment write-ups, analysis logs). Five years as an Environmental Scientist (Utah Division of Oil, Gas & Mining) in geospatial automation and technical presentation.

Why ready now. The work begins with public-archive data acquisition and pipeline construction — the FI's demonstrated strength across both the STV DEM research and the Appalachian project. The new elements (Antarctic hydrology, hierarchical modeling) are scheduled as Year-1 coursework and PI mentorship, before the analyses that need them.

Graduate study timeline. PhD, Geosensing Systems Engineering & Sciences; enrolled August 2025; qualifying/candidacy exam [completed date / planned term]; proposed FINESST period 01/2027–12/2029 (start + 3 years); expected graduation August 2029 [confirm — a full 3-year award through 12/2029 needs graduation ≥ 12/2029]. Milestones: Yr 1 coursework + attribution manuscript; Yr 2 generalization paper + AGU; Yr 3 synthesis + defense.